Probability HW3

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Problem 1

Consider $X \sim Bi(n,p)$ and $Y \sim Bi(m,p)$ independent Z-valued random variables. $P(X+Y=k)$ will be:

$$P(X+Y=k) = P(X=j) \cdot P(Y=k-j)$$

$$P(X+Y=k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j} \binom{m}{k-j} p^{k-j} (1-p)^{m-k+j}$$

$$= p^k (1-p)^{n+m-k} \sum_{j=0}^{k} \binom{n}{j} \binom{m}{k-j} = \binom{n+m}{k} p^k (1-p)^{n+m-k}$$

Divide both sides by $p^k(1-p)^{n+m-k}$ we will have:

$$\sum_{j=0}^{k} \binom{n}{j} \binom{m}{k-j} = \binom{m+n}{k}$$

For example: You have n men and m women, and you need to choose from them a team of size k. How can you choose it? You can choose j men and $k-j$ women and iterate all values of men.

Problem 2

The probability that out of these n balls exactly r were white is given by the **binomial distribution**. The binomial distribution is a discrete probability distribution that gives the probability of r successes in n independent trials, each of which has two possible outcomes (success or failure). In this case, the success is getting a white ball, and the failure is getting a black ball.

The probability of success on each trial is $p = \frac{m_1}{m}$, and the probability of failure is $q = \frac{m_2}{m}$. The binomial distribution for this problem is given by:

$$P(r) = \binom{n}{r} \cdot (p)^r \cdot (q)^{n-r} \quad = \binom{n}{r} \cdot \left(\frac{m_1}{m}\right)^r \cdot \left(\frac{m_2}{m}\right)^{n-r}$$

where:

- P(r) is the probability of getting exactly r white balls in n trials
- n is the number of trials
- r is the number of white balls
- p is the probability of getting a white ball on each trial
- q is the probability of getting a black ball on each trial
- $\binom{n}{r}$ is the binomial coefficient, which is the number of ways to choose r objects from a set of n objects

For example, if $m = 10$, $m_1 = 4$, $m_2 = 6$, $n = 5$, and $r = 3$, then the probability of getting exactly 3 white balls in 5 trials is:

$$P(3) = \binom{5}{3} \cdot \left(\frac{4}{10}\right)^3 \cdot \left(\frac{6}{10}\right)^2 = 0.214990848$$

Problem 3

Just like in problem 1, we derived the formula as below:

$$\begin{aligned}
P(X+Y=r) &= P(X=j) \cdot P(Y=r-j) \\
P(X+Y=r) &= \sum_j \binom{n}{j} p^j (1-p)^{n-j} \binom{m}{r-j} p^{r-j} (1-p)^{m-r+j} \\
&= p^r (1-p)^{n+m-r} \sum_j \binom{n}{j} \binom{m}{r-j} \\
&= \binom{n+m}{r} p^r (1-p)^{n+m-r} \\
&= Z \sim Bi(n+m, p)
\end{aligned}$$

However, the name of this distribution is binomial distribution

Problem 4(i)

To determine the optimal winning strategy for the second player, let's consider different cases based on the value of p_1 chosen by the first player.

Case 1: If p_1 is very close to 0 (e.g., $p_1 < 0.5$):

In this case, the first player's number is very close to 0, and the second player should choose p_2 to be very close to 1. The second player wants to maximize the probability of the first player's number becoming zero while keeping their own number as close to 1 as possible. So, p_2 should be close to 1.

Case 2: If p_1 is close to 0.5:

In this case, p_1 is roughly equal to 0.5, so there's an equal chance of the first player's number becoming zero. The second player should also choose p_2

to be close to 0.5 to have a balanced chance of winning. If p_2 is too close to 1, the second player might lose if the first player's number becomes zero, and vice versa if p_2 is too close to 0.

Case 3: If p_1 is very close to 1 (e.g., $p_1 > 0.5$):

In this case, the first player's number is very close to 1, and the second player should choose p_2 to be very close to 0. The second player wants to maximize the probability of the first player's number becoming zero while keeping their own number as close to 0 as possible. So, p_2 should be close to 0.

In summary, the optimal winning strategy for the second player is as follows:

- If $p_1 < 0.5$, choose p_2 close to 1.
- If $p_1 \approx 0.5$, choose p_2 close to 0.5.
- If $p_1 > 0.5$, choose p_2 close to 0.

Problem 4(ii)

The optimal winning strategy for player one, when player two follows the strategy outlined in the previous response, is to maximize their own winning probability. To do this, player one should choose p_1 in such a way that they exploit player two's strategy and aim to win the game.

Player one's optimal strategy depends on player two's choice of p_2 based on the value of p_1:

1. If player two chooses p_2 close to 1 (following the strategy for Case 1), player one should choose p_1 very close to 0 to maximize their chance of winning. In this case, player two is likely to lose the game because their number is likely to become zero.

2. If player two chooses p_2 close to 0.5 (following the strategy for Case 2), player one should choose p_1 close to 0.5 to balance the chances. This will make it a more even game, and the outcome will depend on random chance.

3. If player two chooses p_2 close to 0 (following the strategy for Case 3), player one should choose p_1 close to 1 to maximize their chance of winning. In this case, player two is likely to lose the game because their number is likely to become zero.

In summary, player one's optimal strategy is to choose p_1 to exploit player two's strategy by maximizing their own winning probability based on the value of p_2 chosen by player two.

Problem 5

The random variable X represents the number of failures in a series of Bernoulli trials with success probability p, carried out until there are r successes. This is known as a Negative Binomial distribution. The Probability Mass Function

(PMF) of a Negative Binomial distribution is given by:

$$P(X = k) = \binom{k + r - 1}{k} \cdot (1 - p)^k \cdot (p)^r$$

Where

- **k** is the number of failures
- **r** is the number of successes
- **p** is the probability of success

This PMF gives you the probability of observing **k** failures before obtaining **r** successes in a series of Bernoulli trials with probability of success **p**.

Problem 6

Find below in the next page an image of the programming and the output to problem 6.

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import nbinom

# Define the number of successes (r)
r = 5

# Define a range of values for the success probability (p)
p_values = [0.2, 0.4, 0.6, 0.8]

# Define a range of values for k (number of failures)
k_values = np.arange(0, 21)

# Create a plot for each value of p
for p in p_values:
    pmf = nbinom.pmf(k_values, r, p)
    plt.plot(k_values, pmf, label=f'p={p}')

# Add labels and a legend
plt.xlabel('Number of Failures (k)')
plt.ylabel('PMF')
plt.title('Negative Binomial PMF for Different Values of p')
plt.legend()
plt.grid(True)

# Show the plot
plt.show()
```

Figure 1: Problem 6 coding

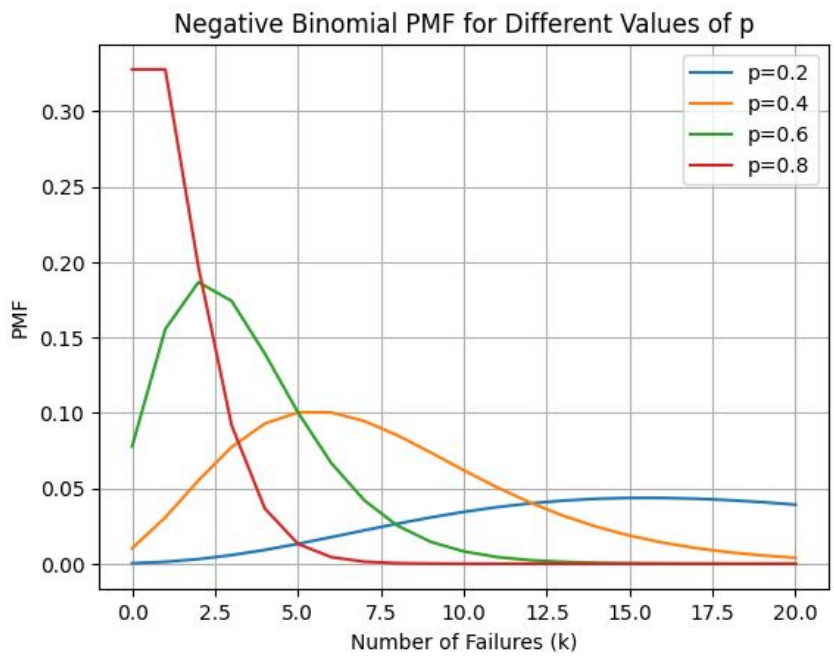

Figure 2: Output for different values of p